

TUTORIAL 1 (FP2)**CHAPTER 1: MATRICES 2****Section I – Notes and Exercises****1.1: SYSTEMS OF LINEAR EQUATIONS****Linear Equations**

1. Recall that in two dimensions a line in a rectangular xy -coordinate system can be represented by an equation of the form

$$ax + by = c \quad (a, b \text{ not both } 0)$$

and in three dimensions a plane in a rectangular xyz -coordinate system can be represented by an equation of the form

$$ax + by + cz = d \quad (a, b, c \text{ not all } 0)$$

These are examples of “linear equations”, the first being a linear equation in the variables x and y and the second a linear equation in the variables x , y , and z .

2. More generally, we define a **linear equation** in the n variables $x_1, x_2, \dots, x_n$ to be one that can be expressed in the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where $a_1, a_2, \dots, a_n$ and b are constants, and **the a 's are not all zero**.

In the special cases where $n = 2$ or $n = 3$, we will often use variables without subscripts and write linear equations as

$$\begin{aligned} a_1x + a_2y &= b \quad (a_1, a_2 \text{ not both } 0) \\ a_1x + a_2y + a_3z &= b \quad (a_1, a_2, a_3 \text{ not all } 0) \end{aligned}$$

3. A finite set of linear equations is called a **system of linear equations** or, more briefly, a **linear system**. The variables are called **unknowns**.

For example, system (1) that follows has unknowns x and y , and system (2) has unknowns x_1 , x_2 , and x_3 .

$$\begin{array}{ll} 5x + y = 3 & 4x_1 - x_2 + 3x_3 = -1 \\ 2x - y = 4 & 3x_1 + x_2 + 9x_3 = -4 \\ \text{System (1)} & \text{System (2)} \end{array}$$

4. A general linear system of m equations in the n unknowns $x_1, x_2, \dots, x_n$ can be written as

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \dots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \dots & + & a_{2n}x_n & = & b_2 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \dots & + & a_{mn}x_n & = & b_m \end{array}$$

5. A **solution** of a linear system in n unknowns $x_1, x_2, \dots, x_n$ is a sequence of n numbers $s_1, s_2, \dots, s_n$ for which the substitution

$$x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$$

makes each equation a true statement.

For example, the system in (1) has the solution

$$x = 1, y = -2$$

and the system in (2) has the solution

$$x_1 = 1, x_2 = 2, x_3 = -1.$$

These solutions can be written more succinctly as

$$(1, -2) \text{ and } (1, 2, -1)$$

in which the names of the variables are omitted. This notation allows us to interpret these solutions **geometrically** as **points** in **two-dimensional** and **three-dimensional space**.

6. More generally, a solution

$$x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$$

of a linear system in n unknowns can be written as

$$(s_1, s_2, \dots, s_n)$$

which is called an **ordered n -tuple**. With this notation it is understood that all variables appear in the same order in each equation. If $n = 2$, then the n -tuple is called an **ordered pair**, and if $n = 3$, then it is called an **ordered triple**.

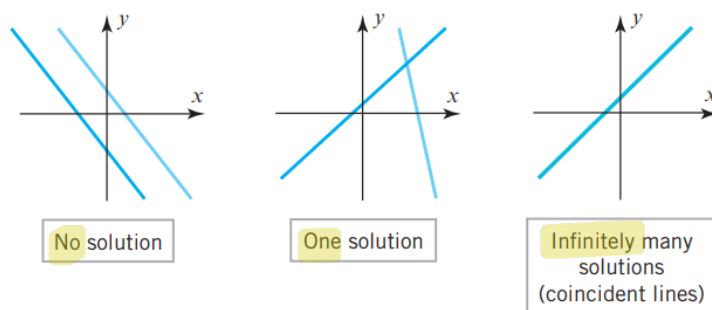
Linear Systems in Two and Three Unknowns

- Linear systems in **two unknowns** arise in connection with **intersections of lines**.
- For example, consider the linear system

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

in which the graphs of the equations are **lines in the xy -plane**. Each **solution** (x, y) of this system corresponds to a **point of intersection** of the lines, so there are three possibilities:



- The **lines** may be **parallel and distinct**, in which case there is **no intersection** and consequently **no solution**.
- The **lines** may **intersect at only one point**, in which case the system has **exactly one solution**.
- The **lines** may **coincide**, in which case there are **infinitely many points of intersection** (the points on the common line) and consequently **infinitely many solutions**.

Beuz 2 straight lines can only intersect once

- Exactly 1
- Infinitely many

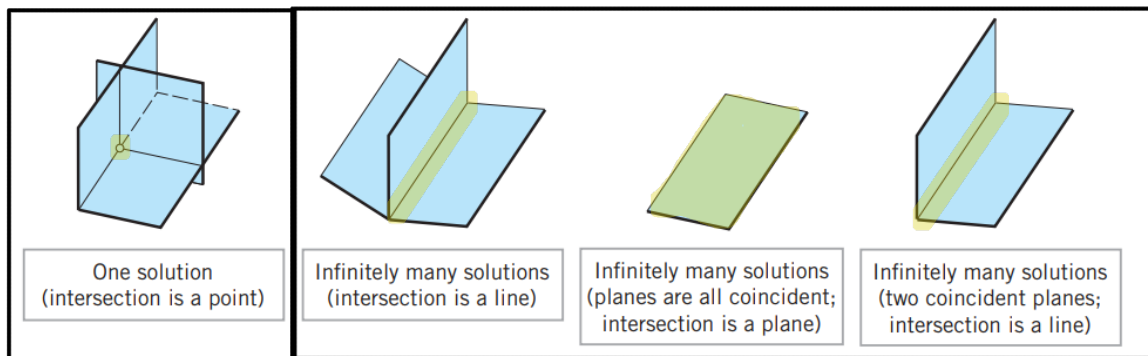
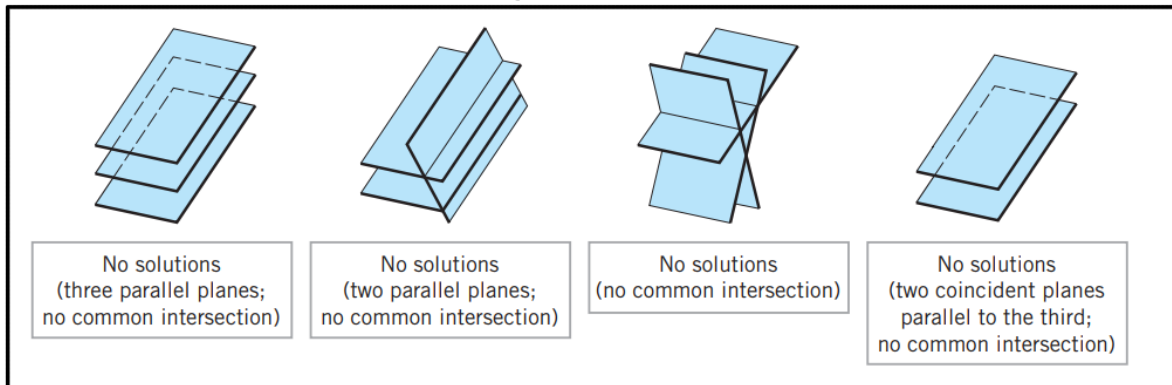
Prepared by AwYeongZS

- In general, we say that a linear system is **consistent** if it has **at least one solution** and **inconsistent** if it has no solutions. Thus, a **consistent** linear system of two equations in two unknowns has either one solution or infinitely many solutions—there are no other possibilities.
- The same is true for a linear system of **three equations in three unknowns**

$$\begin{aligned}a_1x + b_1y + c_1z &= d_1 \\a_2x + b_2y + c_2z &= d_2 \\a_3x + b_3y + c_3z &= d_3\end{aligned}$$

in which the graphs of the equations are **planes**. The **solutions** of the system, if any, correspond to **points where all three planes intersect**, so again we see that there are only three possibilities—no solutions, one solution, or infinitely many solutions.

No solution



one solution

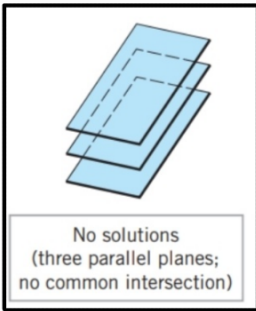
Infinitely many solution

- Our observations about the number of solutions of linear systems of two equations in two unknowns and linear systems of three equations in three unknowns actually hold for *all* linear systems. That is:

Every system of linear equations has zero, one, or infinitely many solutions. There are no other possibilities.

No solutions

1

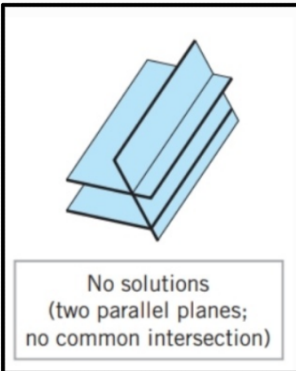


• There are 3 distinct parallel planes.

Example:

$$\begin{cases} x + 2y - 4z = 1 \\ x + 2y - 4z = 5 \\ 2x + 4y - 8z = -3 \end{cases} \quad \begin{cases} \vec{n}_1 = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \\ \vec{n}_2 = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \\ \vec{n}_3 = \begin{pmatrix} 2 \\ 4 \\ -8 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \end{cases}$$

2



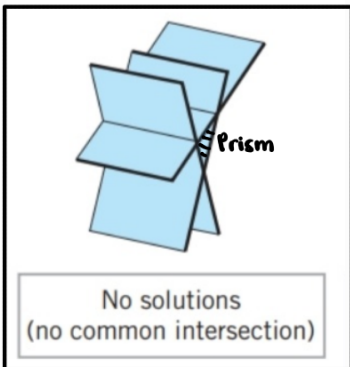
• There are 2 distinct parallel planes.

• The third plane is not parallel to the first two planes

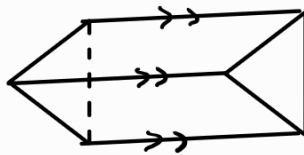
Example:

$$\begin{cases} x + 2y - 4z = 1 \\ x + 2y - 4z = 5 \\ 2x + 4y - 5z = -3 \end{cases} \quad \begin{cases} \vec{n}_1 = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \\ \vec{n}_2 = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \\ \vec{n}_3 = \begin{pmatrix} 2 \\ 4 \\ -5 \end{pmatrix} \end{cases}$$

3

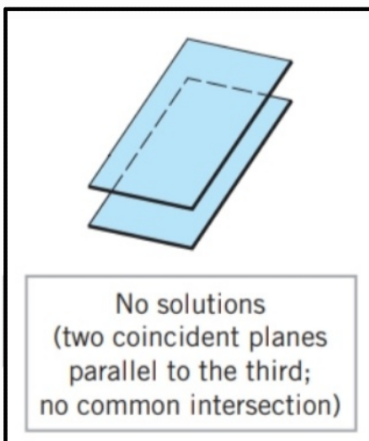


• The 3 planes form a prism



All 3 planes not parallel

4



• There are 2 coincident planes.

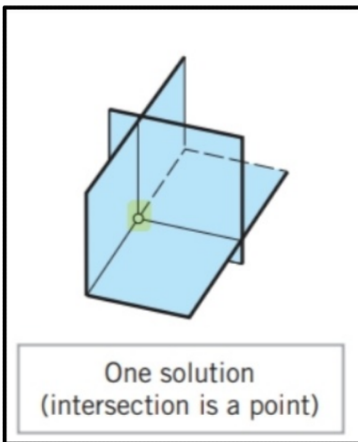
• The third plane is parallel to but distinct from the first two

Example:

$$\begin{cases} x + 2y - 4z = 1 \\ x + 2y - 4z = 1 \\ 2x + 4y - 8z = -3 \end{cases} \quad \begin{cases} \vec{n}_1 = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \\ \vec{n}_2 = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \\ \vec{n}_3 = \begin{pmatrix} 2 \\ 4 \\ -8 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \end{cases}$$

Solution(s)

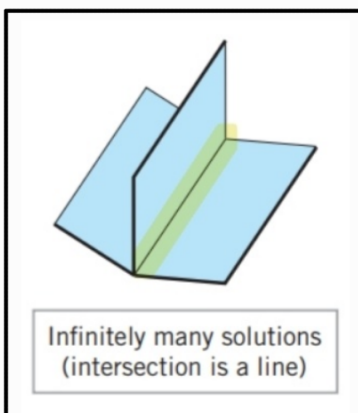
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- The 3 planes intersect at one point

Planes doesn't have to be perpendicular

6



- 3 planes have a line in common

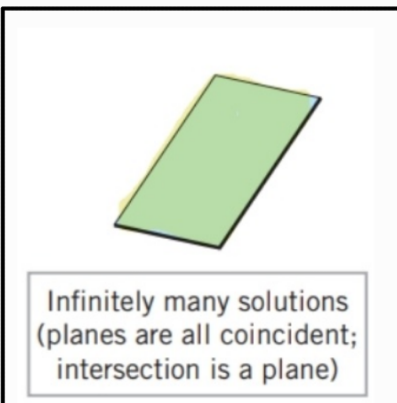
- No 2 planes are parallel

OR

- The 3 planes form a sheaf
[Recommended]

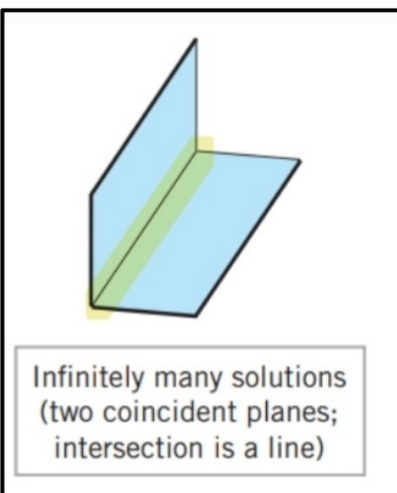


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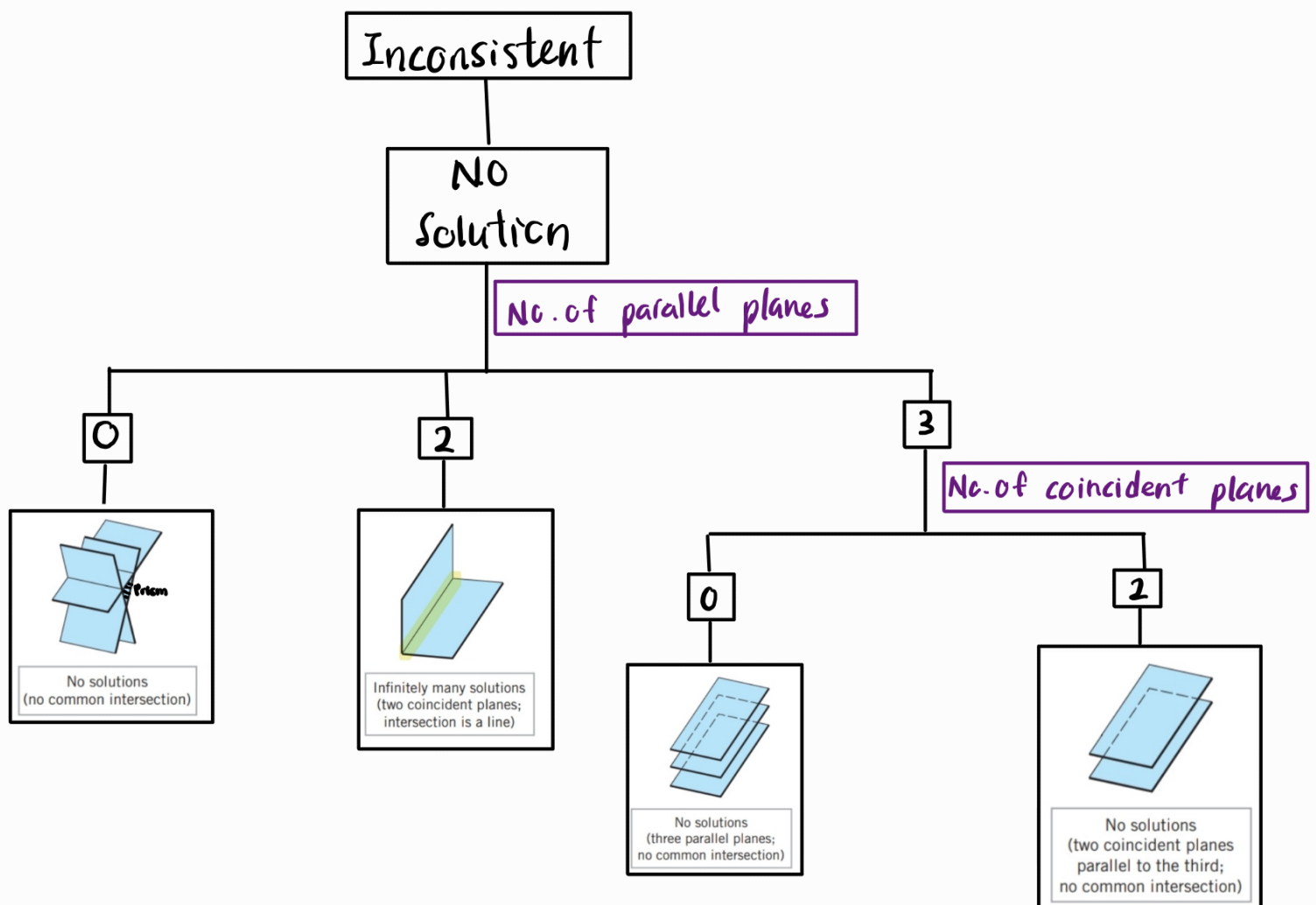
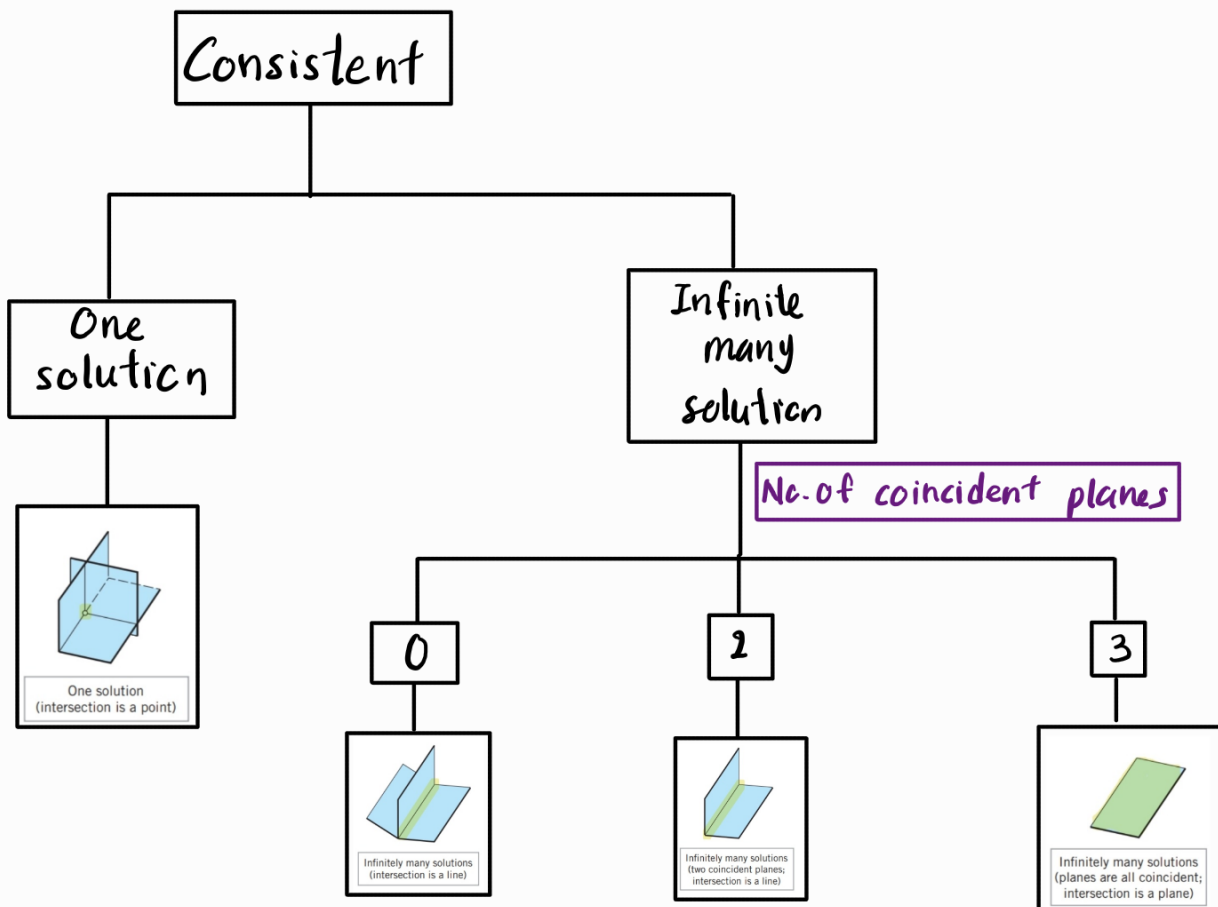


- 3 planes are coincident

8



- 2 coincident planes intersect
the third plane in a line



Rules : ① Only can add eqn

Illustration 1 (A linear system with one solution)

Solve the linear system

$$\begin{aligned}x - y &= 1 \\2x + y &= 6\end{aligned}$$

Discussion

We can eliminate x from the second equation by adding -2 times the first equation to the second. This yields the simplified system

$$\begin{aligned}x - y &= 1 \\3y &= 4\end{aligned}$$

From the second equation we obtain $y = \frac{4}{3}$, and on substituting this value in the first equation we obtain

$$x = 1 + y = \frac{7}{3}.$$

Thus, the system has the **unique solution**

$$x = \frac{7}{3}, y = \frac{4}{3}.$$

Geometrically, this means that the lines represented by the equations in the system **intersect at the single point** $\left(\frac{7}{3}, \frac{4}{3}\right)$.

Activity: Try to check this by graphing the lines.

Illustration 2 (A linear system with no solutions)

Solve the linear system

$$\begin{aligned}x + y &= 4 \\3x + 3y &= 6\end{aligned}$$

Discussion

We can eliminate x from the second equation by adding -3 times the first equation to the second equation. This yields the simplified system

$$\begin{aligned}x + y &= 4 \\0 &= -6\end{aligned}$$

The **second equation** is **contradictory**, so the given system has **no solution**.

Geometrically, this means that the lines corresponding to the equations in the original system are **parallel and distinct**.

Activity: Try to check this by graphing the lines.

Illustration 3 (A linear system with infinitely many solutions)

Solve the linear system

$$\begin{aligned}4x - 2y &= 1 \\16x - 8y &= 4\end{aligned}$$

Discussion

We can eliminate x from the second equation by adding -4 times the first equation to the second. This yields the simplified system

$$\begin{aligned}4x - 2y &= 1 \\0 &= 0\end{aligned}$$

The second equation does not impose any restrictions on x and y and hence can be omitted. Thus, the solutions of the system are those values of x and y that **satisfy the single equation**

$$4x - 2y = 1$$

Geometrically, this means the lines corresponding to the two equations in the original system **coincide**.

One way to describe the solution set is to solve this equation for x in terms of y to obtain $x = \frac{1}{4} + \frac{1}{2}y$ and then assign an arbitrary value t (called a **parameter**) to y . This allows us to **express the solution** by the pair of equations (called **parametric equations**)

$$x = \frac{1}{4} + \frac{1}{2}t, \quad y = t, \quad \text{where } t \in \mathbb{R}$$

We can obtain **specific numerical solutions** from these equations by **substituting numerical values** for the parameter t . For example,

$$t = 0 \text{ yields the solution } \left(\frac{1}{4}, 0\right),$$

$$t = 1 \text{ yields the solution } \left(\frac{3}{4}, 1\right), \text{ and}$$

$$t = -1 \text{ yields the solution } \left(-\frac{1}{4}, -1\right).$$

You can confirm that these are solutions by substituting their coordinates into the given equations.

Note:

1. If a linear system has infinitely many solutions, then a set of parametric equations from which all solutions can be obtained by assigning numerical values to the parameters is called a **general solution** of the system.
2. The solution to the linear system can also be written in the following form:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{4} \\ 0 \end{pmatrix} + t \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix}, \text{ where } t \in \mathbb{R}$$

Illustration 4 (A linear system with infinitely many solutions)

Solve the linear system

$$\begin{aligned}x - y + 2z &= 5 \\2x - 2y + 4z &= 10 \\3x - 3y + 6z &= 15\end{aligned}$$

Discussion

This system can be solved by inspection, since the **second and third equations are multiples of the first**.

Geometrically, this means that **the three planes coincide** and that those values of x, y , and z that satisfy the equation

$$x - y + 2z = 5$$

automatically satisfy all three equations. Thus, it suffices to find the solutions of $x - y + 2z = 5$.

We can do this by first solving this equation for x in terms of y and z , then **assigning arbitrary values r and s (parameters)** to these two variables, and then expressing the solution by the three parametric equations

$$x = 5 + r - 2s, \quad y = r, \quad z = s$$

Specific solutions can be obtained by **choosing numerical values for the parameters r and s** .

For example, taking $r = 1$ and $s = 0$ yields the solution $(6, 1, 0)$.

Note:

The solution to the linear system can also be written in the following form:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} + r \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}, \text{ where } r, s \in \mathbb{R}$$

Illustration 1 (A linear system with one solution)

Solve the linear system

$$\begin{array}{rcl} x - y = 1 & \text{---} & \textcircled{1} \\ 2x + y = 6 & \text{---} & \textcircled{2} \end{array}$$

Discussion

We can eliminate x from the second equation by adding -2 times the first equation to the second. This yields the simplified system

$$\begin{array}{rcl} x - y = 1 \\ 3y = 4 \end{array}$$

$\textcircled{1} \times (-2)$
 $-2x + 2y = -2 \text{ --- } \textcircled{3}$

$\textcircled{3} + \textcircled{2}, 3y = 4$

Original system

$$\begin{cases} x - y = 1 \\ 2x + y = 6 \end{cases}$$

copy

Updated system

$$\begin{cases} x - y = 1 \\ 3y = 4 \end{cases}$$

Use [modification of
eqn ①] + [eqn ②]

Note!

You can choose either one eqn to copy

Solution(s) in both
system will be the
same following this
operation

Illustration 2 (A linear system with no solutions)

Solve the linear system

$$\begin{aligned}x + y &= 4 \\ 3x + 3y &= 6\end{aligned}$$

Discussion

We can eliminate x from the second equation by adding -3 times the first equation to the second equation. This yields the simplified system

$$\begin{aligned}x + y &= 4 \\ 0 &= -6\end{aligned}$$

Original system

$$\begin{cases} x + y = 4 \\ 3x + 3y = 6 \end{cases}$$

$$\begin{aligned} & \textcircled{1} \times (-3), \\ & -3x - 3y = -12 \text{ --- } \textcircled{3} \\ & \textcircled{3} + \textcircled{2} \\ & \hline & 0 = -6 \end{aligned}$$

Updated system

$$\begin{cases} x + y = 4 \\ 0 = -6 \text{ (contradiction)} \end{cases}$$

$\therefore$ No solution

Illustration 3 (A linear system with infinitely many solutions)

Solve the linear system

$$\begin{aligned}4x - 2y &= 1 \\ 16x - 8y &= 4\end{aligned}$$

Discussion

We can eliminate x from the second equation by adding -4 times the first equation to the second. This yields the simplified system

$$\begin{aligned}4x - 2y &= 1 \\ 0 &= 0\end{aligned}$$

Original system

$$\begin{cases} 4x - 2y = 1 \\ 16x - 8y = 4 \end{cases}$$

$$\begin{aligned} & \textcircled{1} \times (-4), \\ & -16x + 8y = -4 \text{ --- } \textcircled{3} \\ & \textcircled{3} + \textcircled{2} \\ & \hline & 0 = 0 \end{aligned}$$

Updated system

$$\begin{cases} 4x - 2y = 1 \\ 0 = 0 \end{cases}$$

Ans

$$\begin{aligned}4x - 2y &= 1 \\ x &= \frac{1}{4} + \frac{1}{2}y\end{aligned}$$

Let $y = t$, where $t \in \mathbb{R}$

Then, $x = \frac{1}{4} + \frac{1}{2}t$

$\therefore$ Soln of system is

$$x = \frac{1}{4} + \frac{1}{2}t, y = t, t \in \mathbb{R}$$

ALT way to present final ans:

$$\begin{aligned} (x, y) &= \left(\frac{1}{4} + \frac{1}{2}t, t\right), t \in \mathbb{R} \\ \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \frac{1}{4} + \frac{1}{2}t \\ t \end{pmatrix} = \begin{pmatrix} \frac{1}{4} \\ 0 \end{pmatrix} + t \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{4} \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \end{pmatrix}, t \in \mathbb{R} \\ &= \begin{pmatrix} \frac{1}{4} \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \end{pmatrix}, t \in \mathbb{R} \end{aligned}$$

Vec Eqn of a line

General Solution

$\therefore$ Correct, but doesn't impose any restrictions
Hence, we can omit eqn $\textcircled{2}$

$\therefore 4x - 2y = 1$ contains all solutions of the updated system

Hence, $4x - 2y = 1$ contains all solutions of the original system

Illustration 4 (A linear system with infinitely many solutions)

Solve the linear system

$$\left. \begin{aligned} x - y + 2z &= 5 \\ 2x - 2y + 4z &= 10 \\ 3x - 3y + 6z &= 15 \end{aligned} \right\} \text{coincident planes}$$

Discussion

This system can be solved by inspection, since the **second and third equations are multiples of the first**.

Original system

$$\begin{cases} x - y + 2z = 5 \\ 2x - 2y + 4z = 10 \\ 3x - 3y + 6z = 15 \end{cases}$$

$$\textcircled{1} \times (-2)$$

$$-2x + 2y - 4z = -10 \text{ --- } \textcircled{3}$$

$$\textcircled{3} + \textcircled{2}$$

$$0 = 0$$

$$\textcircled{1} \times (-3)$$

$$-3x + 3y - 6z = -15 \text{ --- } \textcircled{4}$$

$$\textcircled{4} + \textcircled{3}$$

$$0 = 0$$

Updated system

$$\begin{cases} x - y + 2z = 5 \\ 0 = 0 \\ 0 = 0 \end{cases}$$

$\therefore$ Correct, but doesn't impose any restrictions
Hence, we can omit eqn $\textcircled{2}$ & $\textcircled{3}$

$\therefore x - y + 2z = 5$ contains all solutions of the updated system

Hence, $x - y + 2z = 5$ contains all solutions of the original system

Ans :

$$x - y + 2z = 5$$

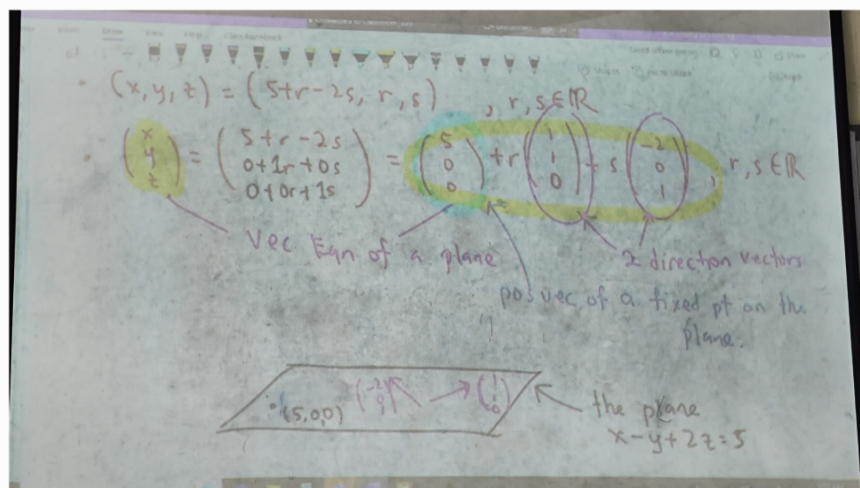
$$x = 5 + y - 2z$$

Let $y = r$, where $r \in \mathbb{R}$

$z = s$, where $s \in \mathbb{R}$

Then, $x = 5 + r - 2s$

$\therefore$ Soln is $x = 5 + r - 2s$, $y = r$, $z = s$, where $r, s \in \mathbb{R}$



Solving a System of Linear Equations Using Elementary Row Operations (ERO)

1. The system of linear equations

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$a_{21}x + a_{22}y + a_{23}z = b_2$$

$$a_{31}x + a_{32}y + a_{33}z = b_3$$

can be written as

$$\begin{matrix} \text{Matrix of} \\ \text{coefficients} \end{matrix} \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \quad \begin{matrix} \text{column vector} \\ \mathbf{x} \end{matrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{\vec{x}} \quad \mathbf{\vec{x}} = \mathbf{\vec{b}}$$

Denoting $\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$, $\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$, we now have a matrix equation $\mathbf{Ax} = \mathbf{b}$.

Here, $\mathbf{A}$ is called the **coefficient matrix** (or **matrix of coefficients**) of the system.

2. An **augmented matrix** of a system consists of the **coefficient matrix** with an **added column** containing the **constants** from the right sides of the equations.

The **augmented matrix** of the system above is

$$\left(\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array} \right)$$

$$\begin{array}{l} \text{Original} \quad \left\{ \begin{array}{l} x - y = 1 \\ 2x + y = 6 \end{array} \right. \end{array}$$

$$\begin{array}{l} \text{Replacement} \rightarrow \left\{ \begin{array}{l} x - y = 1 \\ 3y = 4 \end{array} \right. \\ \text{Interchange} \rightarrow \left\{ \begin{array}{l} 2x + y = 6 \\ x - y = 1 \end{array} \right. \\ \text{Scaling} \rightarrow \left\{ \begin{array}{l} 10x - 10y = 10 \\ 2x + y = 6 \end{array} \right. \end{array}$$

$$\text{Updated}$$

3. Three basic operations are used to simplify a linear system:

- replace** one equation by the **sum of itself and a multiple of another equation**,
- interchange** two equations, and
- multiply** all the terms in an equation by a **nonzero constant**.

$$\text{Augmented matrix} \quad \left(\begin{array}{cc|c} 1 & -1 & 1 \\ 2 & 1 & 6 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 + (-2R_1)} \left(\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 3 & 4 \end{array} \right)$$

4. Operations on equations in a linear system correspond to **operations on the appropriate rows of the augmented matrix**. The three basic operations listed earlier correspond to the following operations on the augmented matrix.

Elementary Row Operations (ERO):

- (a) (Replacement) **Replace** one row by the **sum of itself and a multiple of another row**.

$$R_i \rightarrow R_i + kR_j$$

- (b) (Interchange) **Interchange** two rows.

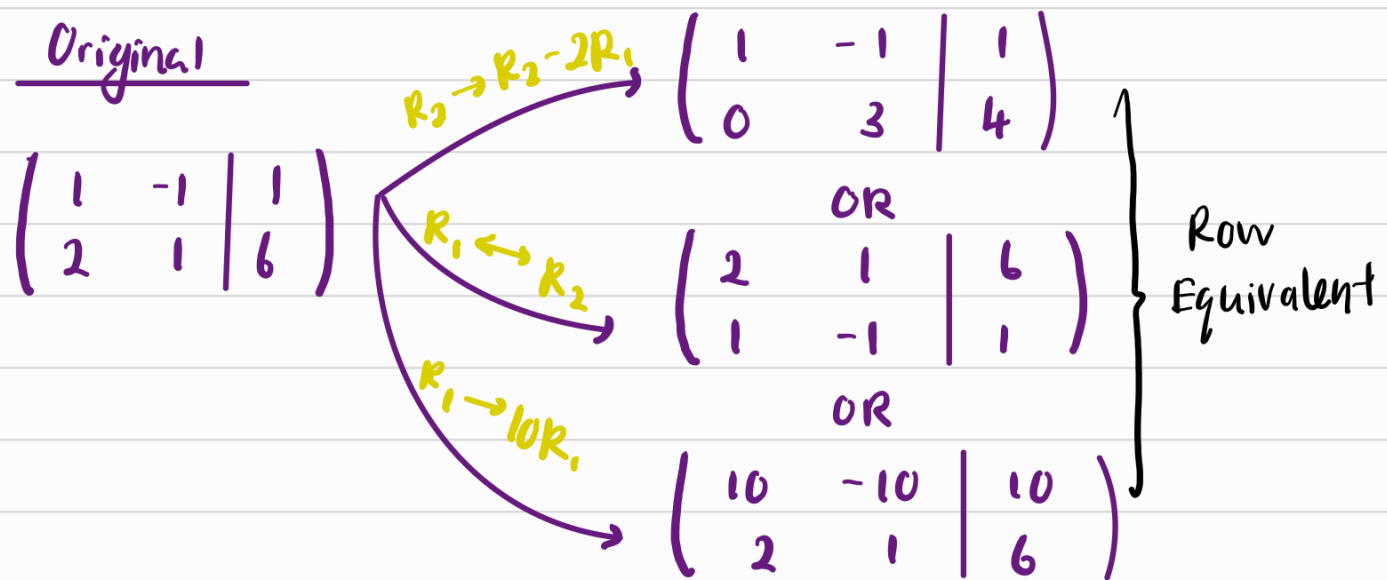
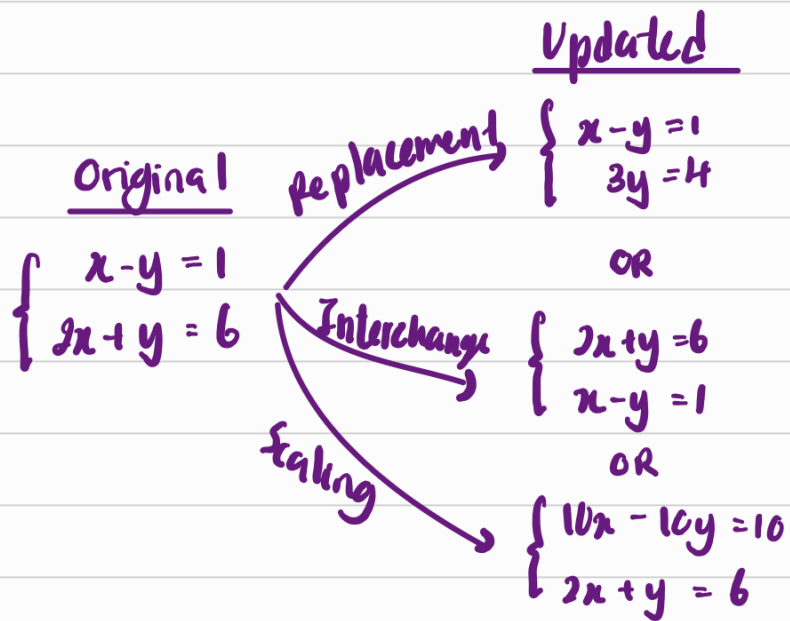
$$R_i \leftrightarrow R_j$$

- (c) (Scaling) **Multiply** all entries in a row by a **nonzero constant**.

$$R_i \rightarrow kR_i$$

5. Row operations can be applied to any matrix, not merely to one that arises as the augmented matrix of a linear system. Two matrices are called **row equivalent** if there is a sequence of elementary row operations that transforms one matrix into the other.

[**Note:** Row operations are **reversible**.]



6. At the moment, we are interested in row operations on the augmented matrix of a system of linear equations. Suppose a system is changed to a new one via row operations. By considering each type of row operation, you can see that any solution of the original system remains a solution of the new system. Conversely, since the original system can be produced via row operations on the new system, each solution of the new system is also a solution of the original system. This discussion justifies the following statement.

If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.

Example 1

Solve the following system of linear equations using elementary row operations (ERO).

$$x + y + 2z = 9$$

$$2x + 4y - 3z = 1$$

$$3x + 6y - 5z = 0$$

(Ans: $x = 1, y = 2, z = 3$)

Make sure top left is +1

Solution

Make these 0

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array} \right] \xrightarrow{\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1}} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array} \right]$$

Make this +1

$$\xrightarrow{R_2 \rightarrow \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 3 & -11 & -27 \end{array} \right]$$

make this 0

$$\xrightarrow{R_3 \rightarrow R_3 - 3R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & -1/2 & -3/2 \end{array} \right]$$

Make this +1

$$\xrightarrow{R_3 \rightarrow -2R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Make these 0

$$\xrightarrow{\substack{R_1 \rightarrow R_1 - 2R_3 \\ R_2 \rightarrow R_2 + \frac{7}{2}R_3}} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Make this 0

$$\xrightarrow{R_1 \rightarrow R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Gaussian Elimination
completed
[REF]

$\therefore$ Solution of system is
 $x = 1, y = 2, z = 3$

Gauss-Jordan Elimination
completed [RREF]

Echelon Forms

1. In Example 1, we solved a linear system in the unknowns x , y , and z by reducing the augmented matrix to the form

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

from which the solution $x = 1$, $y = 2$, $z = 3$ became evident. This is an example of a matrix that is in **reduced row echelon form (RREF)**. To be of this form, a matrix must have the following properties:

causes only zero to be at the bottom of leading 1

- If a row does not consist entirely of zeros, then the first nonzero number in the row is a 1. We call this a **leading 1**.
- If there are any rows that consist entirely of zeros, then they are grouped together at the bottom of the matrix.
- In any two successive rows that do not consist entirely of zeros, the **leading 1 in the lower row** occurs **farther to the right** than the leading 1 in the higher row.
- Each column that contains a leading 1 has **zeros everywhere else in that column**.

A matrix that has the **first three properties** [(a), (b) and (c)] is said to be in **row echelon form (REF)**. (Thus, a matrix in reduced row echelon form is of necessity in row echelon form, but not conversely.)

2. Examples:

The following matrices are in reduced row echelon form (RREF).

$$\begin{array}{c} \text{a)} \end{array} \begin{array}{c} \text{d)} \end{array} \left[\begin{array}{cccc} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{array} \right], \quad \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right], \quad \begin{array}{c} \text{c)} \end{array} \begin{array}{c} \text{d)} \end{array} \left[\begin{array}{cccccc} 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right], \quad \left[\begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array} \right]$$

The following matrices are in row echelon form (REF) but *not* reduced row echelon form (RREF).

$$\left[\begin{array}{cccc} 1 & 4 & -3 & 7 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 1 & 5 \end{array} \right], \quad \left[\begin{array}{ccc} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right], \quad \left[\begin{array}{ccccc} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right]$$

3. As the examples in point #2 above illustrate, a matrix in row echelon form (REF) has zeros below each leading 1, whereas a matrix in reduced row echelon form (RREF) has zeros below *and above* each leading 1. Thus, with any real numbers substituted for the *'s, all matrices of the following types are in row echelon form (REF):

$$\left[\begin{array}{cccc} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 1 \end{array} \right], \quad \left[\begin{array}{cccc} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{array} \right], \quad \left[\begin{array}{cccc} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right], \quad \left[\begin{array}{cccccccc} 0 & 1 & * & * & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 0 & 1 & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{array} \right]$$

All matrices of the following types are in reduced row echelon form (RREF):

$$\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right], \quad \left[\begin{array}{cccc} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{array} \right], \quad \left[\begin{array}{cccc} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right], \quad \left[\begin{array}{cccccccc} 0 & 1 & * & 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{array} \right]$$

4. The procedure (or algorithm) we have just done (in Example 1) for reducing a matrix to reduced row echelon form (RREF) is called **Gauss–Jordan elimination**. This algorithm consists of two parts, a **forward phase** in which zeros are introduced below the leading 1's and a **backward phase** in which zeros are introduced above the leading 1's.

If only the forward phase is used, then the procedure produces a row echelon form (REF) and is called **Gaussian elimination**.

5. For small linear systems, Gauss–Jordan elimination (reduction to RREF) is a good procedure to use.

However, it is generally **more efficient** to use **Gaussian elimination** (reduction to REF) followed by a technique known as **back-substitution** to complete the process of solving the system. Example 2 below illustrates this technique.

Example 2

Solve Example 1 again, this time using Gaussian elimination, followed by back-substitution.

$$x + y + 2z = 9$$

$$2x + 4y - 3z = 1$$

$$3x + 6y - 5z = 0$$

$$(Ans: x = 1, y = 2, z = 3)$$

Solution

Make sure top left is +1

Make these 0

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 - 3R_1]{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array} \right]$$

Make this +1

$$\xrightarrow{R_2 \rightarrow \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 3 & -11 & -27 \end{array} \right]$$

make this 0

$$\xrightarrow{R_3 \rightarrow R_3 - 3R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} \end{array} \right]$$

Make this +1

$$\xrightarrow{R_3 \rightarrow -2R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Make these 0

Gaussian Elimination completed

[REF]

Back Sub

$$z = 3$$

$$x + y + 2z = 9 \quad \text{--- ①}$$

$$y - \frac{7}{2}z = -\frac{17}{2} \quad \text{--- ②}$$

$$\text{Sub } z = 3 \rightarrow \text{②}$$

$$y - \frac{7}{2}(3) = -\frac{17}{2}$$

$$y = 2$$

$$\therefore x + 2 + 2(3) = 9$$

$$x = 1$$

Example 3

Find the unique solution for the system of equations

$$2x + 5z = -3$$

$$x + y + 2z = 0$$

$$x - y + 4z = -4$$

(Ans: $x = 1, y = 1, z = -1$)

Solution

$$\left[\begin{array}{ccc|c} 2 & 0 & 5 & -3 \\ 1 & 1 & 2 & 0 \\ 1 & -1 & 4 & -4 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 2 & 0 & 5 & -3 \\ 1 & -1 & 4 & -4 \end{array} \right]$$

$$\begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array} \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & -2 & 1 & -3 \\ 0 & -2 & 2 & -4 \end{array} \right]$$

Example 4

Find a solution set for the system of equations

$$2x + 3y + z = 1$$

$$4x + 10y + 5z = 3$$

$$2x + 11y + 7z = 3$$

$$\left(\text{Ans: } x = \frac{1}{8} + \frac{5}{8}t, y = \frac{1}{4} - \frac{3}{4}t, z = t, t \in \mathbb{R} \right)$$

Solution

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 4 & 10 & 5 & 3 \\ 2 & 11 & 7 & 3 \end{array} \right] \xrightarrow[\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1}]{} \left[\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 0 & 4 & 3 & 1 \\ 0 & 8 & 6 & 2 \end{array} \right]$$

$$\xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \left[\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 4 & 3 & 1 \\ 0 & 8 & 6 & 2 \end{array} \right]$$

$$\xrightarrow{R_3 \rightarrow R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 4 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\xrightarrow{R_2 \rightarrow \frac{1}{4}R_2} \left[\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & \frac{3}{4} & \frac{1}{4} \\ 0 & 0 & 0 & 0 \end{array} \right]$$

REF

Back sub

$$y + \frac{3}{4}z = \frac{1}{4}$$

$$y = \frac{1}{4} - \frac{3}{4}z$$

$$\text{Let } \boxed{z = t, t \in \mathbb{R}}$$

$$y = \frac{1}{4} - \frac{3}{4}t$$

$$\begin{aligned} x + \frac{3}{2}y + \frac{1}{2}z &= \frac{1}{2} \implies x = \frac{1}{2} - \frac{3}{2}y - \frac{1}{2}z \\ &= \frac{1}{2} - \frac{3}{2}\left(\frac{1}{4} - \frac{3}{4}t\right) - \frac{1}{2}(t) \\ &= \frac{1}{2} - \frac{3}{8} + \frac{9}{8}t - \frac{1}{2}t \\ &= \frac{1}{8} + \frac{5}{8}t \end{aligned}$$

$$\therefore \boxed{x = \frac{1}{8} + \frac{5}{8}t, y = \frac{1}{4} - \frac{3}{4}t, z = t, t \in \mathbb{R}}$$

Example 5

Show that the system of equations

$$x + 4y - 2z = 1$$

$$x + 5y = 2$$

$$3x + 13y - 4z = 3$$

has no solutions.

Solution

$$\left[\begin{array}{ccc|c} 1 & 4 & -2 & 1 \\ 1 & 5 & 0 & 2 \\ 3 & 13 & -4 & 3 \end{array} \right] \xrightarrow{\substack{R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 3R_1}} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 0 \end{array} \right]$$

$$\xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & -1 \end{array} \right]$$

$\therefore$ 3rd row gives $0z = -1$, which is a contradiction.

$\therefore$ The system of eqns has no solutions.

Provided that any other row is non-zero
and is not a contradiction

Note:

For a system of equations, where the REF of the augmented matrix is of the form $\left(\begin{array}{ccc|c} * & * & * & * \\ * & * & * & * \\ 0 & 0 & \alpha & \beta \end{array} \right)$, there are

3 cases to consider:

(a) If $\alpha = 0$ and $\beta = 0$, then there is an **infinite number of solutions**.

(b) If $\alpha = 0$ and $\beta \neq 0$, then there are **no solutions**.

(c) If $\alpha \neq 0$, then for all $\beta \in \mathbb{R}$ there will be a **unique solution**.

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 1 & 5 & 8 \\ 0 & 0 & \alpha & \beta \end{array} \right)$$